1. Prove that ${}_{k|}q_x = {}_kp_x\, q_{x+k}$

2. Z is the present value random variable for a 15-year pure endowment of 1 on (x):

   (i) The force of mortality is constant over the 15-year period.

   (ii) v = 0.9

   (iii) Var(Z)= 0.065E[Z]

   Calculate q_x

3. For a whole life insurance of 1000 on (x) with benefits payable at the moment of death:

   (i) The force of interest at time t, $\delta_t = \begin{cases} 0.04, & 0 < t \leq 10 \\ 0.05, & 10 < t \end{cases}$

   (ii) $\mu_{x+t} = \begin{cases} 0.06, & 0 < t \leq 10 \\ 0.07, & 10 < t \end{cases}$

   Calculate the single benefit premium for this insurance.

4. For a special whole life insurance on (x), payable at the moment of death:

   (i) $\mu_{x+t} = 0.05, \quad t > 0$

   (ii) $\mu = 0.08$

   (iii) The death benefit at time t is $b_t = e^{0.06t}, t > 0.$

   (iv) Z is the present value random variable for this insurance at issue.

   Calculate Var (Z) .

5. For (x):

   (i) K is the curtate future lifetime random variable

   (ii) $q_{x+k} = 0.1(k+1), \quad k = 0, 1, 2, \ldots, 9$

   (iii) X= min(K,3)

   Calculate Var(X)

6. Based on a 4% annual interest rate, evaluate

   (a) ${}_5E_{40}$

   (b) D_{35}

   (c) $A^{\ 1}_{25:\overline{10|}}$

7. For a special fully discrete 3-year term insurance on (x):

   (i) The death benefit payable at the end of year k+1 is
   $$b_{k+x} = \begin{cases} 0 & for\ k = 0 \\ 1000(11-k) & for\ k = 1,2 \end{cases}$$

   (ii) The mortality probalities are given as:

k	q_{k+1}
0	0.200
1	0.100
2	0.097

   (iii) $i = 0.06$

   Calculate the level annual benefit premium for this insurance.

8. For a special 3-year term insurance on (x), you are given:

   (i) Z is the present-value random variable for the death benefits.

   (ii) $q_{x+k} = 0.02(k+1) \quad k = 0, 1, 2$

   (iii) The following death benefits, payable at the end of the year of death:

k	b_{k+1}
0	300,000
1	350,000
2	400,000

   (iv) $i = 0.06$

9. A maintenance contract on a hotel promises to replace burned out light bulbs at the end of each year for three years. The hotel has 10,000 light bulbs. The light bulbs are all new. If a replacement bulb burns out, it too will be replaced with a new bulb.
   You are given:

   (i) For new light bulbs, $q_0 = 0.10$, $q_1 = 0.30$, $q_2 = 0.50$

   (ii) Each light bulb costs 1.

   (iii) $i = 0.05$

   Calculate the actuarial present value of this contract.

10. A whole life insurance of 1 issued to (20) is payable at the moment of death. You are given:

    (i) The time-at-death random variable is exponential with $\mu = 0.02$

(ii) $\delta = 0.14$

Calculate $\bar{A}_{20}, {}^2\bar{A}_{20}$ and $Var(\bar{Z}_{20})$

11. You are given the following information:

    (i) A unit benefit n-year term life insurance policy.

    (ii) The age-at-death random variable is uniform on $[0, \omega]$

    (iii) The constant force of interest δ

    (iv) $\bar{Z}^{\ 1}_{x:\overline{n}|}$ is the contingent payment random variable for a life aged x.

    Find $\bar{A}^{\ 1}_{x:\overline{n}|}$, ${}^2\bar{A}^{\ 1}_{x:\overline{n}|}$ and $Var(\bar{Z}^{\ 1}_{x:\overline{n}|})$

12. The age-at-death random variable is exponential with parameter 0.05. A life aged 35 buys a 25-year life insurance policy that pays 1 upon death. Assume an annual effective interest rate of 7%, find the actuarial present value of this policy.

13. The lifetime of a group of people has the following survival function associated with it: $s(x) = 1 - x/100$, $0 \le x \le 100$. Paul, a member of the group, is currently 40 years old and has a 15-year endowment insurance policy, which will pay him $50,000 upon death. Find the actuarial present value and standard deviation of this policy. Assume an annual force of interest $\delta = 0.05$

14. You are given the following survival function for a newborn:
    $S_0(t) = \frac{(121-t)^{1/2}}{k}, \quad t \in [0, \omega]$

    (a) Show that k must be 11 for $S_0(t)$ to be a valid survival function.

    (b) Show that the limiting age, ω, for this survival model is 121.

    (c) Calculate $\mathring{e}_5$ for this survival model.

    (d) Derive an expression for μ_x for this survival model, simplifying the expression as much as possible.

    (e) Calculate the probability, using the above survival model, that (57) dies between the ages of 84 and 100.

15. Let the remaining lifetime at birth random variable be uniform on [0,250]. let Z be the contingent payment random variable for a life aged x=30. Find $A^{\ 1}_{30:\overline{10}|}$, ${}^2A^{\ 1}_{30:\overline{10}|}$ and Var(Z) if $\delta = 0.05$

16. Let the remaining lifetime at birth random variable be uniform on [0,100]. let Z_{30} be the contingent payment random variable for a life aged x=30. Find $A_{30}, {}^2A_{30}$ and $Var(Z_{30})$ if $\delta = 0.05$

17. Let the remaining lifetime at birth random variable be uniform on [0,120]. let ${}_{10|}Z_{30}$ be the contingent payment random variable for a life aged x=30. Find ${}_{10|}A_{30}$, ${}_{10|}^{2}A_{30}$ and $Var({}_{10|}Z_{30})$ if $\delta = 0.06$

18. Let the remaining lifetime at birth random variable X be exponential with $\mu = 0.06$. Let $Z^1_{30:\overline{10}|}$ be the contingent payment random variable for a life aged x=30.
Find $A^1_{30:\overline{10}|}$, ${}^2A^1_{30:\overline{10}|}$ and $Var(Z^1_{30:\overline{10}|})$ if $\delta = 0.10$

19. Let the remaining lifetime at birth random variable X be exponential with $\mu = 0.06$. Let $Z_{30:\overline{10}|}$ be the contingent payment random variable for a life aged x=30.
Find $A_{30:\overline{10}|}$, ${}^2A_{30:\overline{10}|}$ and $Var(Z_{30:\overline{10}|})$ if $\delta = 0.10$